PROBABILITY

- **Calculate the probability using R script for each of the following situations:**

(a) A standard normally distributed variable is larger than 3.

(b) A normally distributed variable with mean 35 and standard deviation 6 is larger than 42.

(c) Getting 10 out of 10 successes in a binomial distribution with probability 0.8.

INFIDELITY

- Research has looked at emotional reactions to infidelity and found that men get homicidal and suicidal and women feel undesirable and insecure.
- Let's imagine we did some similar research: we took some men and women and got their partners to tell them they had slept with someone else. We then took each person to two shooting galleries and each time gave them a gun and 100 bullets. In one gallery was a human-shaped target with a picture of their own face on it, and in the other was a target with their partner's face on it. They were left alone with each target for 5 minutes and the number of bullets used was measured.

Load: **Infidelity.csv**

- Load the saved data.
- Plot a clustered error bar on the mean number of bullets used against self and partner for males and females, separately.

BOOK PUBLISHER

- One book publisher decided to design an experiment in which participants read their book (*Women are from Mars*) and one of the other publisher (*Field's statbook*) at different times. Happiness was measured after reading each book. They collected a sample of 500 participants, but got each participant to take part in reading both books. The order in which books were read was counterbalanced and there was a delay of 6 months between reading the books. They predicted that reading their wonderful books would lead to greater happiness than reading Field's tedious statistics book.

Load:
FieldBook.dat

- Load the dataset.
- Plot the happiness level corresponding to the two publishers.
- Perform the appropriate t-test on the comparison and report the effect size.